

GREENTECH MANUFACTURING OPERATIONS INITIATIVE

Optimising Production Schedules to Reduce Downtime

Using SQL and Power BI to Support Data-Driven Decision Making

PREPARED BY

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TECHNICAL STACK

SQL Server & Power BI

ANALYSIS PERIOD

August 2025

| About GreenTech Manufacturing

Delivering sustainable consumer goods since 2010.

GreenTech produces key environment-friendly products:

- ✓ **Eco-Cleaning:** GreenFoam Soap, BioWipes, EcoWash
- ✓ **Recyclable Packaging:** RePack Packaging Film
- ✓ **Appliances:** Energy-Efficient consumer range

THE IMMEDIATE CHALLENGE

Production delays caused by downtime are directly reducing factory efficiency and negatively affecting customer delivery times.



| Why Was This Project Needed?

Operational Bottlenecks

- **Frequent Machine Failures:** Unplanned maintenance spikes.
- **Material & Scheduling Conflicts:** Ineffective supply timing.
- **Sanitation Overhead:** Lengthy cleaning activities block runs.
- **Manual Processes:** Vulnerable to human planning errors.

This leads directly to delayed customer orders, reduced overall factory output, and highly inflated operational expenses.

ESTIMATED LOSSES

\$1.5M

Annualised Financial Cost

Losses due to unnecessary downtime, planning delays, and lost throughput capacity.

| Project Aim

“To analyse production and downtime data to pinpoint the causes of operational delays and deliver scheduling recommendations that enhance efficiency, cut back on downtime, and lift GreenTech's productivity.”



| Key Objectives



1. Root Causes

Determine and classify the primary drivers contributing to frequent downtime events across lines.



2. Products

Isolate and identify the key products experiencing high frequencies of batch-level production delays.



3. Operators

Analyse patterns across operator shifts to distinguish training needs from systemic operational challenges.



4. Scheduling

Assess overlaps, bottleneck dependencies, and conflicts occurring from manual planning processes.



5. Dashboards

Build and deliver high-visibility Power BI visual reporting dashboards for operational stakeholders.



6. Action Plan

Formulate data-backed recommendations to establish a baseline for continuous operational improvement.

| Business Questions Addressed



Operational Downtime

What causes delays most frequently? Which specific bottleneck types result in the longest cumulative production delays?



Product Impact

Which items suffer the highest occurrences of downtime? Which product categories incur the highest loss of productive run-hours?



Operator Analysis

Do specific operators exhibit higher delays, or are the root causes primarily independent of team member actions?



Scheduling Integrity

Are manual production plans causing over-scheduling and direct operational conflicts on the plant floor?

| Data Scope & Transformation

Underlying Datasets

Dataset	Purpose / Scope
Products	Master records, dimensions, standard run plans.
Line Productivity	Production batches completed, operator assignments.
Line Downtime	Timestamp logs of operational halts and disruptions.
Downtime Factors	Standard classification keys for root-cause codes.

Workflow Infrastructure

- SQL** **Extraction & Cleansing**
Used to standardise datatypes, handle null values, write query transformations, and compute core runtime metrics.
- PBI** **Business Intelligence Modeling**
Built a robust, relational star-schema model to enable dynamic visualizations, slicers, and performance analysis.

| Our Analysis Methodology

1

Understanding

Map factory workflow, key metrics, and data structures.

2

Cleansing

Handle structural transformations via optimized SQL views.

3

Downtime

Assess patterns, frequency counts, and root factors.

4

Products

Pinpoint high-risk lines and actual vs planned variance.

5

Operators

Review workload alignment and delay patterns.

6

Scheduling

Examine manual overlapping runs and floor conflicts.

7

BI Tools

Deploy intuitive interactive reporting dashboards.

8

Actions

Draft and propose high-impact operational guidelines.

Executive Performance Overview

645

BATCHES RUN

363

DELAYED
BATCHES

21.7

TOTAL DAYS LOST

885

DOWNTIME
EVENTS

13

DISTINCT CAUSES

4

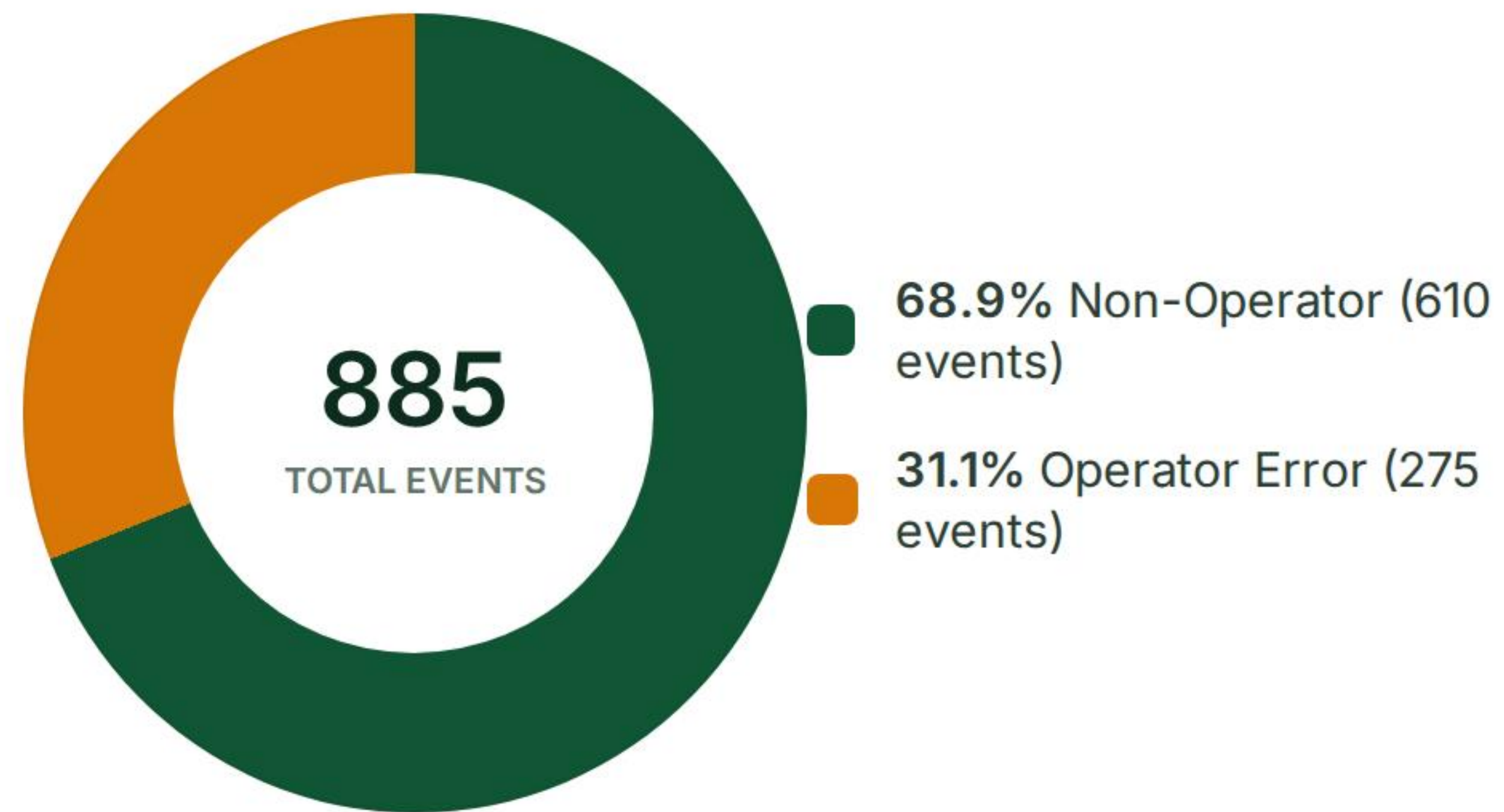
MAX CUTS PER
RUN

Critical Insight

56.2% of all production batches completed experienced at least one operational delay.

This confirms that production delays are a systematic norm rather than exception, signaling a severe need for structured process intervention.

| Process vs Operator Root Causes



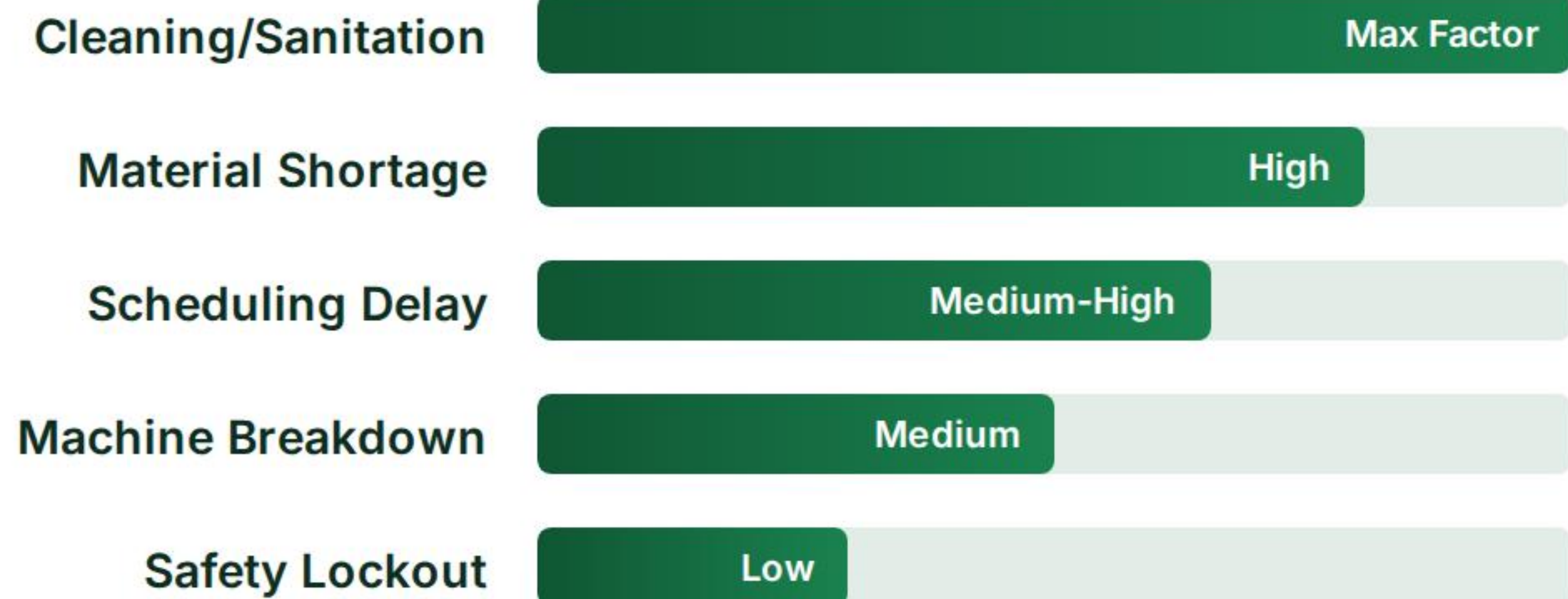
Systemic Process Flaws

Over two-thirds of operational halts stem directly from broader process errors, inventory lapses, or machine breakdowns rather than operator mistakes.

STRATEGIC IMPLICATIONS

Remediation should target operational protocols, material pipeline predictability, and preventative equipment checks rather than personnel monitoring.

| Primary Causes of Downtime



Downtime Driver Breakdown

Cleaning and Sanitation processes emerged as the single highest source of recurring operational delays.

- ✓ Indicates that clean-up windows are not optimized or integrated effectively into production run schedules.
- ✓ Raw material constraints and scheduling friction outpace physical machine failures as delay drivers.

| Most Vulnerable Products

Product-Specific Vulnerability

GreenFoam Hand Soap exhibits the highest rate of production interruptions across all lines.

- ✓ GreenFoam line recorded **323 distinct downtime events**.
- ✓ BioWipes followed at **204 events**.
- ✓ RePack Film (185) and EcoWash (173) show relatively lower but still substantial disruption rates.

Targeting optimization efforts on the GreenFoam line will yield the highest cumulative efficiency gains.



| Planned vs Actual Run Times

Run-Time Excess Analysis

Across all product lines, actual run durations consistently exceed planned parameters, indicating systemic schedule slippage.

Key Takeaway

Schedules are written based on optimistic, frictionless run-rates rather than realistic, historical performance windows.

Product Name	Planned (Hrs)	Actual (Hrs)	Variance (%)
EcoWash Liquid	8.00	8.91	+11.38%
RePack Film	8.00	8.87	+10.87%
BioWipe Sheets	7.00	7.72	+10.28%
GreenFoam Soap	6.00	6.79	+13.17%

| Operator Performance Dynamics

Data Breakdown

Highest Absolute Downtime Frequency:

Paul | James | Emily

Highest Percentage of Delayed Batches:

- ✓ Linda (70.73% of runs experienced delay)
- ✓ Sophia (65.00% of runs experienced delay)
- ✓ Rita (63.41% of runs experienced delay)

⚠ Context Is Key

This variance does **not** necessarily indicate poor operator capability. Rather, performance metrics are heavily influenced by:

- ✓ Complexity of specific assigned lines (e.g., GreenFoam)
- ✓ Imbalanced workload distribution
- ✓ Systemic scheduling errors that occur during certain shifts

| Manual Scheduling Inefficiencies

Manual Planning Overhead

Lack of coordination from manual planning generates direct operational conflicts on the plant floor.

- Multiple instances where a single operator was assigned to distinct, overlapping runs simultaneously.
- Lack of visibility results in material shortages colliding with peak production runs.

OVERLAP BOTTLENECK

51 Days

With Overlapping Runs Scheduled

High-conflict scheduling directly creates bottlenecks and line idle times.

| Summary of Core Findings

⚙️ 1. Systemic Process Issues

69% of halts are process-driven. It is an operational flow and equipment issue, not an operator compliance problem.

🧹 2. Clean-Up Overhead

Sanitation procedures represent the single most frequent delay. Standardised run times ignore these intervals.

🧴 3. Soap Line Critical

GreenFoam Hand Soap line suffers worst from delays. Resolving constraints here yields the highest impact.

⚠️ 4. Overlap & Conflict

51 production days suffered from overlapping manual schedules, causing clear execution friction.

📅 5. Variance Slippage

Actual runs consistently outlast standard planning windows by up to 13%, causing downstream delays.

🏭 6. Supply Chain Gap

Material shortages outpace machine issues as delay factors, pointing to poor inventory alignment.

| Actionable Recommendations

1. Inventory Alerts

Introduce automated low-stock alerts and safety thresholds inside SQL/ERP systems to prevent raw material pauses.

2. Sanitize Optimization

Optimize clean-up intervals and build structured, realistic sanitization blocks directly into the daily master run plan.

3. Preventive PMs

Shift from reactive to predictive maintenance scheduling on high-failure production lines to reduce sudden halts.

4. Automated Schedulers

Replace manual whiteboards with interactive, data-driven Power BI / ERP scheduling tools to detect overlap conflicts.

5. Load Balancing

Implement smart workload distribution rules based on line complexity to avoid overloading specific operators.

6. Monthly Reviews

Adopt continuous integration loops with monthly operational Power BI downtime audits to trace target adherence.

| Expected Business Benefits

Efficiency Gains

- ✓ **Reduced Downtime:** Targeted intervention in sanitation and scheduling can recoup up to 30% of lost hours.
- ✓ **Greater Output:** Smooth transitions between runs translates to higher cumulative weekly batches.
- ✓ **Reduced Expenses:** Avoiding overruns limits team overtime pay and reactive support expenses.

Strategic and Team Gains

- ✓ **On-Time Delivery:** Streamlined scheduling ensures accurate, predictable delivery to end customers.
- ✓ **Operator Morale:** Transparent, optimized load distributions reduce work stress and friction.
- ✓ **Data-First Culture:** Management benefits from clean, accessible visual analytics in real time.

| Limitations of the Analysis

Analysis Constraints

While providing clear operational direction, the underlying findings must be interpreted alongside key system boundaries:

- ✓ Data coverage is restricted to a 6-month period (March to August 2025).
- ✓ The analysis is heavily reliant on the precision of manual logs completed on the floor.
- ✓ Wider supply-chain friction (e.g., vendor-level dispatch lags) was not assessed.

⚠ Analytical Note

Operational improvements based on this data assume stable raw material availability and steady demand. Unchecked external vendor variables could offset anticipated efficiency gains.

STRATEGIC SUMMARY

Unlocking Factory Potential Through Data

Downtime at GreenTech is driven predominantly by operational and process constraints rather than individual operator performance. Focusing on system automated tools, predictive maintenance, and strategic inventory buffers will reliably reduce delays, improve productivity, and support profitable green growth.

DATA SYSTEM
SQL Views

REPORTING TOOL
Power BI

| Image Sources



https://www.foodengineeringmag.com/ext/resources/eNews/2020/FE_Enews-SusPkging_EUAutomation.webp?t=1603897052

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